

Race, Gender, and Wages in High-Skill Occupations: An Intersectional Analysis

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I. Introduction

This research project investigates whether Black women in high-skill occupations face an intersectional wage penalty that exceeds the combined effects of race and gender penalties experienced separately by Black men and white women. Wage differentials are estimated using data from the IPUMS-ACS (American Community Survey) for the years 2015-2019, focusing on full-time workers aged 25 to 64 employed in professional, managerial, technical, and other high-skill occupations. This paper contributes to the literature by providing a quantitative test of intersectionality within high-skill professions, using representative ACS data and a structured sequence of regression models.

The following empirical approach consists of estimating log-wage regressions to compare annual wage and salary income across demographic groups: White men, White women, Black men, and Black women, both without and after controlling for variables such as age, potential work experience, years of education, detailed occupation, industry, and state of residence. To test for intersectionality, it is assessed whether the wage penalty for Black women significantly exceeds the sum of the penalties for being Black and for being female. This is accomplished by including an interaction term (Black $\times$ Female) in the regression model to identify non-additive, intersectional effects. A negative and statistically significant coefficient on this interaction term

would provide evidence that Black women face unique labor market discrimination beyond what can be explained by race and gender penalties separately.

II. Literature Review

Research in labor economics has documented persistent wage disparities by race and gender. Studies find that Black workers earn less than White workers (Bayer & Charles, 2018), and women earn less than men even when controlling for education and work experience (Blau & Kahn, 2017). However, traditional analyses often consider these identities separately. Intersectionality, a concept introduced by Crenshaw (1989), argues that identities interact in ways that produce qualitatively different forms of disadvantage. Recent empirical research, including McLaughlin and Neumark (2024), explores whether wage discrimination operates additively or whether certain groups, especially Black women, face compounded penalties. According to the Institute for Women's Policy Research, there is a national statistic that black women earn about 63-69 cents less than white men.

Existing evidence on intersectional wage discrimination is mixed. Some studies report disproportionately large penalties for Black women, while others find that selection into higher-skill occupations leads to milder penalties than predicted by an additive model. This paper extends this literature using ACS microdata and a structured modeling approach.

Despite the developments we have made as a society, wage inequality in high-skill occupations continues, and this is particularly troubling because jobs where meritocracy is supposed to prevail, such as lawyers, doctors, engineers, corporate executives, and technology professionals, operate in environments that emphasize credentials, performance metrics, and objective qualifications. Despite these fields being highly merit-based, we continue to see wage

gaps. It is studied that Black workers earn less than their white counterparts, and women earn less than men, but it is important to note that the gap is lessening with the furthering of education for minority groups.

III. Background on the Data

For the data used in this project, the source is IPUMS-ACS. This dataset has a large sample size, enabling us to analyze small subgroups, such as black women in high-skill jobs. This is a nationally representative sample, with reliable annual income data and occupation codes to identify high-skill positions. This analysis covers the period 2015-2019, as wages were affected by the pandemic in 2020, which could bias our findings due to pandemic-related external factors. Also, combining 5 years of data provides greater statistical power and more reliable estimates for the analysis.

Some limitations I imposed on the data include a population age range of 25-64 to ensure educational attainment, full-time workers (defined as working 35+ hours/week, 50+ weeks/year), and positive wage income. Also, "high-skill" workers were identified using IPUMS OCC codes, including managers, professionals, and technical occupations such as lawyers, doctors, engineers, software developers, executives, and scientists. These occupations are identified as high-skilled because they require more years of education to perform. Provided are the OCC Variable Occupation Codes that were used to identify these occupations: Management occupations (codes 0010-0430), Business and financial operations (codes 0500-0950), Computer and mathematical occupations (codes 1000-1240), Architecture and engineering (codes 1300-1560), Life, physical, and social science occupations (codes 1600-1980), Legal occupations (codes 2100-2160), Healthcare practitioners and technical occupations (codes 3000-3540).

In terms of race categories, this analysis focuses on four groups defined by the RACE and SEX variables: White men (the reference group), Black men, White women, and Black women. This restriction to Black and White workers allows for a focused analysis of the Black-White wage gap at the intersection with gender. However, it is acknowledged that other racial and ethnic groups also face significant wage disparities; for this research, the scope of these groups and genders is narrowed.

IV. Key Variables

Below are the key variables I extracted from the IPUMS database for analyzing racial wage disparities in high-occupation industries. Each variable is selected based on its relevance to measuring earnings differences and controlling for factors that may influence the wage outcomes.

1. Outcome variable
 - a. INCWAGE — annual wage/salary income
2. Key predictors
 - a. RACE — to identify Black vs. White
 - b. SEX — to identify Male vs. Female
3. Demographics
 - a. AGE — needed as a control
 - b. STATEFIP — state fixed effects (significant for wage models)
4. Work intensity / full-time filter
 - a. UHRSWORK — hours worked/week
 - b. WKSWORK1 — weeks worked/year
5. Occupation

- a. OCC — needed to select high-skill occupations + include as control
- 6. Weights
 - a. PERWT — person weight required for ACS survey design

V. Possible Restrictions

It is important to note that while IPUMS-ACS provides robust data, it also imposes several restrictions on use and affects interpretation. First, the income data is based on self-reporting, which may introduce measurement error. Also, the dataset does not capture within-occupation variation in job titles, responsibilities, or workplace characteristics that might explain some wage differences. In addition, the analysis cannot account for unobservable factors such as workplace discrimination, bias in performance evaluations, or differential access to mentorship and networking opportunities. Additionally, the classification of high-skill jobs was based on my judgment, which may introduce bias and affect the results. Also, the cross-sectional nature of the data prevents making causal claims about the sources of wage disparities. As this analysis is mainly just of black and white workers, my analysis excludes Hispanic, Asian, Native American, and other racial/ethnic groups. Wage disparities for these groups and potential intersectional effects with gender remain unexamined in this aspect of the project.

VI. Methods

To investigate whether Black women face an intersectional wage penalty in high-skill occupations, I employed a log-wage regression framework that allowed the decomposition of wage differences by race and gender. The dependent variable in all models is the natural logarithm of annual wage and salary income ($\ln\text{wage}$), which I chose because it allows coefficients to be interpreted as approximate percentage differences in wages. The core empirical specification takes the following form:

$$\ln(\text{wage}_i) = \beta_0 + \beta_1 \text{Black}_i + \beta_2 \text{Female}_i + \beta_3 (\text{Black} \times \text{Female})_i + X_i' \gamma + \varepsilon_i$$

Where $\ln(\text{wage}_i)$ is the natural log of annual wage and salary income for individual i , Black_i is an indicator variable equal to 1 if the individual is Black, 0 if White, Female_i is an indicator variable equal to 1 if the individual is female, 0 if male, $(\text{Black} \times \text{Female})_i$ is the interaction term, equal to 1 only for Black women, X_i is a vector of control variables that varies across model specifications, and ε_i is the error term. The coefficients in this specification have clear interpretations where β_1 captures the wage penalty for Black men relative to white men, β_2 captures the wage penalty for white women relative to white men, and β_3 captures the intersectional effect, which is the additional wage penalty that Black women experience beyond what would be predicted by simply adding the race and gender penalties. To test for intersectionality, I examined the interaction coefficient β_3 . So, if β_3 is significantly negative, this indicates that Black women face an intersectional penalty. If β_3 is significantly positive, this means a positive interaction, in which the combined effect is less severe than predicted by the additive model. If β_3 is not substantially different from zero, the additive model holds.

For the analysis, four models were estimated, progressively adding control variables to examine how wage gaps change as we account for different factors. I will discuss these models soon. I used the sequential modeling approach because it demonstrates how wage gaps evolve as controls are added, thereby helping us identify which factors contribute to observed disparities. Additionally, it reveals whether education, occupation, or industry sorting accounts for wage differences. The complete model provides the cleanest test of wage discrimination by comparing workers with similar credentials doing similar jobs in similar industries.

All of the models use robust standard errors to account for heteroskedasticity in the wage distribution. This ensures that hypothesis tests and confidence intervals remain valid even when wage variances differ across demographic groups or income levels. All the regressions use IPUMS person weights (PERWT) to ensure estimates are representative of the U.S. population. The ACS uses a complex survey design, and proper weighting is essential for obtaining nationally representative estimates.

VII. Empirical Results and Models

A. Model 1: Baseline Model (No Controls)

The baseline survey-weighted regression with no control variables reveals substantial wage disparities across all groups. The model includes 1,794,400 observations representing a population size of 178,255,539 workers. Black men face a wage penalty of -31.1% relative to white men, while white women experience a -36.8% penalty.

Most notably, the interaction term for Black women shows a positive coefficient of 0.166. The linear combination test indicates an intersectional effect of +0.845, indicating that Black women earn approximately 84.5% more than would be predicted by summing the Black and female penalties.

This counterintuitive finding in the baseline model highlights the critical importance of including appropriate control variables, as raw wage comparisons do not account for differences in age, education, occupation, or industry composition across demographic groups. The R-squared of 0.0693 indicates that race and gender alone account for less than 7% of the

variation in wages. <Model 1: No Controls>

```
. * MODEL 1: No controls
. svy: reg lnwage i.black##i.female
(running regress on estimation sample)
```

Survey: Linear regression

Number of strata =	1	Number of obs =	1,794,400
Number of PSUs =	1,794,400	Population size =	178,255,539
		Design df =	1,794,399
		F(3, 1794397) =	28030.27
		Prob > F =	0.0000
		R-squared =	0.0693

lnwage	Linearized		t	P> t	[95% conf. interval]	
	Coefficient	std. err.				
1.black	-.3110271	.0039704	-78.34	0.000	-.3188088	-.3032453
1.female	-.3678717	.0013779	-266.97	0.000	-.3705724	-.365171
black#female						
1 1	.1663446	.0048787	34.10	0.000	.1567825	.1759067
_cons	11.25492	.0010293	1.1e+04	0.000	11.2529	11.25694

```
. estimates store m1
. lincom 1.black#1.female - (1.black + 1.female)
( 1) - 1.black - 1.female + 1.black#1.female = 0
```

lnwage	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
(1)	.8452434	.008883	95.15	0.000	.827833	.8626538

B. Model 2: Adding Age Controls

When age and age-squared are added to control for experience and career progression, the patterns shift somewhat. The Black penalty decreases to -29.6%, while the female penalty

remains substantial at -35.6%. The interaction term remains positive at 0.149 ($t = 31.41$, $p < 0.001$).

The age coefficients reveal the expected relationship: earnings increase with age at a decreasing rate, consistent with standard human capital theory. Additionally, the R-squared increases to 0.1239, indicating that age accounts for additional variation in wages beyond race and gender alone. However, the persistence of significant main effects for both race and gender suggests that age differences alone cannot explain the observed wage gaps.

<Model 2: Add Age>

```
. * MODEL 2: Add age
. svy: reg lnwage i.black##i.female c.age c.age2
(running regress on estimation sample)
```

Survey: Linear regression

Number of strata =	1	Number of obs =	1,794,400
Number of PSUs =	1,794,400	Population size =	178,255,539
		Design df =	1,794,399
		F(5, 1794395) =	32404.12
		Prob > F =	0.0000
		R-squared =	0.1239

lnwage	Linearized		t	P> t	[95% conf. interval]	
	Coefficient	std. err.				
1.black	-.2956647	.0038427	-76.94	0.000	-.3031962	-.2881332
1.female	-.3556621	.0013311	-267.19	0.000	-.3582711	-.3530531
black#female						
1 1	.1488662	.0047402	31.41	0.000	.1395757	.1581568
age	.0848633	.0005057	167.80	0.000	.0838721	.0858546
age2	-.000816	5.72e-06	-142.54	0.000	-.0008272	-.0008048
_cons	9.198467	.0106629	862.66	0.000	9.177568	9.219366

```
. estimates store m2

. lincom 1.black#1.female - (1.black + 1.female)

( 1) - 1.black - 1.female + 1.black#1.female = 0
```

lnwage	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
(1)	.800193	.0086034	93.01	0.000	.7833307	.8170553

C. Model 3: Adding Education Controls

Model 3 introduces education controls, which substantially improve the model fit (R-squared increases to 0.2057 from 0.1239). The Black penalty decreases to -26.0%, and the female penalty reduces to -36.3%. The Black×Female interaction term remains positive at 0.152 ($t = 33.42$, $p < 0.001$).

The education coefficients show the expected pattern: higher levels of educational attainment are strongly associated with higher wages. Compared with workers with less than a high school education, workers with higher levels of education exhibit different wage premiums. For example, those with 1 year of college earn approximately 12.5% more, whereas those with 5+ years of college (including graduate degrees) earn approximately 60.2% more.

However, even after accounting for educational differences, which could explain some of the wage gaps if Black women systematically had lower education levels, we still observe large race and gender penalties. The fact that the intersectional interaction term remains positive and significant even after controlling for education suggests that Black women in high-skill occupations may be selected on unobservables: those who successfully enter these professional

fields may possess characteristics (motivation, ability, networks) that partially offset the discrimination they face.

<Model 3: Add Education>

```
. * MODEL 3: Add education
. svy: reg lnwage i.black##i.female c.age c.age2 i.educ
(running regress on estimation sample)
```

Survey: Linear regression

Number of strata =	1	Number of obs =	1,794,400
Number of PSUs =	1,794,400	Population size =	178,255,539
		Design df =	1,794,399
		F(15, 1794385) =	18415.50
		Prob > F =	0.0000
		R-squared =	0.2057

lnwage	Linearized		t	P> t	[95% conf. interval]	
	Coefficient	std. err.				
1.black	-.2601592	.0037179	-69.98	0.000	-.2674461	-.2528723
1.female	-.3629566	.0012697	-285.87	0.000	-.365445	-.3604681
black#female						
1 1	.1519161	.0045462	33.42	0.000	.1430057	.1608266
age	.0821849	.0004867	168.87	0.000	.081231	.0831388
age2	-.000773	5.50e-06	-140.48	0.000	-.0007838	-.0007622
educ						
Nursery school to grade 4	-.0054152	.0430271	-0.13	0.900	-.0897468	.0789164
Grade 5, 6, 7, or 8	-.1436594	.0280822	-5.12	0.000	-.1986996	-.0886193
Grade 9	-.1692294	.0311153	-5.44	0.000	-.2302143	-.1082444
Grade 10	-.0856317	.027235	-3.14	0.002	-.1390115	-.032252
Grade 11	-.1064117	.0248332	-4.29	0.000	-.155084	-.0577395
Grade 12	.0203679	.0197245	1.03	0.302	-.0182914	.0590272
1 year of college	.1253978	.0197425	6.35	0.000	.0867032	.1640924
2 years of college	.1643434	.0197233	8.33	0.000	.1256863	.2030004
4 years of college	.4391565	.019667	22.33	0.000	.4006098	.4777032
5+ years of college	.6015814	.019674	30.58	0.000	.563021	.6401419
_cons	8.857316	.022145	399.97	0.000	8.813912	8.900719

```
. estimates store m3

. lincom 1.black#1.female - (1.black + 1.female)

( 1) - 1.black - 1.female + 1.black#1.female = 0
```

lnwage	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
(1)	.7750319	.0082869	93.52	0.000	.7587898	.791274

D. Model 4: Full Model with Occupation and Industry Controls (MAIN MODEL)

The complete model includes controls for age, education, detailed occupation codes, and industry codes. This is the most rigorous specification, as it compares workers who are as similar as possible across all observable dimensions. The R-squared increases substantially to 0.3628, indicating that occupation and industry account for substantial variation in wages. The model is based on 1,794,400 observations representing 178,255,539 workers, with an F-statistic of 1298.78 ($p < 0.001$).

In this full specification, the Black penalty is -16.5%, the Female penalty is -19.4%, and the Black $\times$ Female interaction is +10.4%. Key findings that can be interpreted through these models are that the total black women penalty is calculated as the combined effect for Black women ($1.\text{black} + 1.\text{female} + 1.\text{black}\#1.\text{female}$), the total wage penalty can be calculated as: $-0.1648 + (-0.1943) + 0.1035 = -0.256$, or approximately -25.7% relative to white men after controlling for age, education, occupation, and industry.

The critical test for intersectionality examines whether Black women face an additional penalty beyond what would be predicted by simply adding the Black penalty and the female penalty. With the interaction term of +0.1036 ($t = 24.30$, $p < 0.001$), we are provided an answer

that if discrimination operated in an additive manner, we would expect Black women to earn 35.9% less than white men (the sum of -16.5% and -19.4%). Instead, Black women earn only 25.7% less, which is a difference of approximately 10.3%.

This positive and highly significant coefficient indicates that Black women earn approximately 10.4% more than would be predicted by simply adding the penalties for being Black and being female. In other words, while Black women do face a substantial wage penalty, this penalty is significantly less than the sum of the separate race and gender penalties.

Once we control for the specific occupations and industries in which workers are employed, Black women do not face an intersectional wage penalty that exceeds the additive effects of race and gender. Instead, they experience a positive interaction effect that substantially reduces the combined penalty relative to what would be expected under a simple additive model. This finding is statistically significant and robust across all model specifications.

<Model 4: Full Model>

```
. * MODEL 4: Add occupation and industry (MAIN MODEL)
. svy: reg lnwage i.black##i.female c.age c.age2 i.educ i.occ i.ind
(running regress on estimation sample)
```

Survey: Linear regression

```
Number of strata =      1          Number of obs = 1,794,400
Number of PSUs   = 1,794,400      Population size = 178,255,539
Design df        = 1,794,399      F(451, 1793949) = 1298.78
Prob > F         = 0.0000         R-squared       = 0.3628
```

lnwage	Linearized					
	Coefficient	std. err.	t	P> t	[95% conf. interval]	
1.black	-.1648312	.0035051	-47.03	0.000	-.1717011	-.1579614
1.female	-.194318	.0013397	-145.04	0.000	-.1969438	-.1916921
black#female						
1 1	.1035527	.0042614	24.30	0.000	.0952006	.1119048
age	.0733163	.0004498	162.98	0.000	.0724347	.074198
age2	-.0006898	5.06e-06	-136.29	0.000	-.0006997	-.0006798
Public finance, taxation, and monetary policy	.0874797	.0184308	4.75	0.000	.051356	.1236035
Administration of human resources programs	.0093818	.0171679	0.55	0.585	-.0242667	.0430302
Administration of environmental quality and housing programs	.0194256	.0181707	1.07	0.285	-.0161884	.0550395
Administration of economic programs	.0952926	.0174563	5.46	0.000	.0610787	.1295064
National security and international affairs	.1911242	.0170574	11.20	0.000	.1576923	.224556
Army	.1052501	.0203747	5.17	0.000	.0653163	.1451839
Air Force	.078411	.0214666	3.65	0.000	.0363372	.1204849
Navy	.1581316	.0224122	7.06	0.000	.1142046	.2020587
Marines	.149632	.0307877	4.86	0.000	.0892893	.2099748
Coast Guard	.1175963	.0339731	3.46	0.001	.0510103	.1841824
Armed Forces, branch not specified	-.0057946	.0802748	-0.07	0.942	-.1631305	.1515413
Military Reserves or National Guard	-.1066246	.0666676	-1.60	0.110	-.2372909	.0240417
_cons	9.569055	.026676	358.71	0.000	9.516771	9.621339

Now, moving on to the critical test of intersectionality, we examine whether Black women face an additional penalty beyond what would be predicted by simply summing the Black men penalty (-16.5%) and the white women penalty (-19.4%). If we look at a purely additive model of discrimination, we would expect having:

$$\text{Black women's penalty} = \text{Black penalty} + \text{Female penalty}$$

$$\text{Expected penalty} = -16.5\% + (-19.2\%) = -35.7\%$$

What we actually observe, though:

$$\text{Black women penalty} = -25.7\%$$

$$\text{Difference} = -25.7\% - (-35.7\%) = \sim +10 \text{ percentage points}$$

To formally test whether this difference is statistically significant, a linear combination test was conducted. The intersectional test yields a coefficient of 47.6%, a t-statistic of 61.35, and a p-value < 0.000. This highly significant positive coefficient suggests that Black women earn approximately 47.6% more than would be predicted by simply adding together the penalties for being Black and being female. While this may seem counterintuitive, it reflects that the actual combined penalty of -25.7% is substantially lower than the additive prediction of -35.7%.

In other words, while Black women still earn substantially less than White men, they do not earn as much less as predicted under a purely additive model. Instead, there is a strong positive interaction effect that reduces the combined penalty. This finding is statistically significant, highly robust, and holds across all model specifications. It suggests that the relationship between race and gender in determining wages is more complex than a simple additive framework can capture.

VII. Discussion and Interpretation

The results of this analysis yield a surprising and policy-relevant finding: Black women in high-skill occupations do not face an intersectional wage penalty that exceeds the sum of separate race- and gender-based penalties. In fact, the interaction term is consistently positive across all model specifications. The formal linear combination test shows that Black women earn 47.6% more than the additive model would predict, meaning their total penalty (-25.7%) is substantially smaller than the sum of the Black men's penalty (-16.5%) and the white women's penalty (-19.4%), which would yield -35.7% under a purely additive framework.

<Model 5: Intersectional Penalty>

```
. lincom 1.black#1.female - (1.black + 1.female)
```

```
( 1)  - 1.black - 1.female + 1.black#1.female = 0
```

lnwage	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
(1)	.4760876	.0077603	61.35	0.000	.4608777	.4912974

```
. display "INTERSECTIONAL penalty: " %4.1f (abs(r(estimate))*100) "%"
INTERSECTIONAL penalty: 47.6%
```

```
. display "P-value: " %5.3f r(p)
P-value: 0.000
```

```
. lincom 1.black + 1.female + 1.black#1.female
```

```
( 1)  1.black + 1.female + 1.black#1.female = 0
```

lnwage	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
(1)	-.2568995	.0026127	-98.33	0.000	-.2620202	-.2517787

```
. display "Total Black women penalty: " %4.1f (abs(r(estimate))*100) "%"
Total Black women penalty: 25.7%
```

```
. lincom 1.black
```

```
( 1)  1.black = 0
```

lnwage	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
(1)	-.1648312	.0035051	-47.03	0.000	-.1717011	-.1579614

```
. display "Black men penalty: " %4.1f (r(estimate)*100) "%"
Black men penalty: -16.5%
```

```
. lincom 1.female
```

```
( 1)  1.female = 0
```

lnwage	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
(1)	-.194318	.0013397	-145.04	0.000	-.1969438	-.1916921

```
. display "White women penalty: " %4.1f (r(estimate)*100) "%"
White women penalty: -19.4%
```

VIII. Policy Recommendations

Based on the finding that we observed where Black women in high-skill occupations face a total wage penalty of about 25.7% relative to white men, some policy recommendations could be implemented. One possible solution is to expand access to high-skill occupations for Black women. This could include targeting scholarships and loan forgiveness for graduate and professional degrees; pipeline programs in STEM and professional fields, such as summer bridge programs, paid internships, and apprenticeships that partner with law firms, tech companies, hospitals, and financial institutions to recruit Black women students into high-skill career tracks; and more vigorous enforcement of Title VII and anti-discrimination laws against discriminatory screening.

Another possible resolution is to strengthen pay transparency and pay equity requirements. Evidence from OECD countries and U.S. states suggests that pay-transparency rules and pay-gap reporting can narrow gender gaps, especially when combined with enforcement and sanctions. Recent state laws (for example, salary-range transparency mandates in multiple states) are explicitly justified as tools to reduce racial and gender wage gaps, and has shown some improvement in that area, which makes it a more effective method to minimize these gaps now.

A final possible resolution would be to tackle occupational and workplace segregation. Research indicates that segregated workplaces, where Black women are clustered in low-status roles or departments, depress their wages by limiting access to lucrative tasks, information networks, and high-pay firms. Occupational segregation accounts for roughly 30 percent of the gender pay gap and is particularly severe for Black women. But, by strengthening enforcements

around promotions, assignments, and access to profit-generating tasks, this gap could also be minimized, and have a better dispersion of workers around various areas in companies.

By centering and prioritizing women of color, policymakers can pinpoint some of the broadest economic disparities, target policy interventions, and lift all boats along the way. These recommendations address some of those areas that could be improved for a more equitable society to develop in the future.

IX. Conclusion

This study finds that Black women in high-skill occupations do not face an intersectional wage penalty that exceeds the sum of penalties for race and gender separately. Instead, conditional on age, education, occupation, and industry, Black women experience a positive interaction effect of 10.4%, meaning their total wage penalty (approximately -25.7%) is substantially smaller than what would be predicted by simply adding the Black men penalty (-16.5%) and white women penalty (-19.4%) together (which would yield -35.9%). This finding is robust and statistically significant across multiple model specifications.

However, this result should not be interpreted as evidence that discrimination against Black women has been eliminated or that intersectionality is irrelevant to labor market outcomes. Instead, it highlights the complexity of how race and gender interact in shaping economic outcomes. The positive interaction effect likely reflects strong selection into high-skill professional occupations: Black women who successfully enter these fields may possess exceptional qualifications and unmeasured abilities that help them overcome structural barriers.

Moreover, wage disparities represent only one dimension of labor market inequality, as women may face discrimination in hiring, promotion, job assignments, and workplace treatment that cannot be measured in cross-sectional wage comparisons, a focus that future research should continue to emphasize. From a policy perspective, these findings suggest that policies or interventions aimed at improving labor market outcomes for Black women should focus not only on wages but also on expanding access for more marginalized communities, supporting career advancement, and addressing workplace discrimination. While wage gaps conditional on employment may be smaller than expected, barriers to entering and advancing in professional fields remain substantial and warrant continued attention from researchers, policymakers, and advocates through greater awareness and further research.

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